

RLHF and data: demographics / nationality

→ the annotator distribution for RLHF can shift its behaviors

→ annotators style matters

⇒ GPT is a surprisingly good pairwise feedback system

⇒ longer effects are a significant outcome of RLHF response

⇒ RLHF and data $\leftrightarrow$ self-training

How do we RLHF?

→ maximize Reward

1. PPO: (Proximal Policy Optimization) = original and very finicky approach

2. ~~PPO~~

trial
step 1: policy gradient update
trial
step 2: TRPO (linearize the problem around current policy)
trial
step 3: PPO (clip the ratios at some eps)

Full RL

→ use Reward model

→ slow/expensive/high variance: sampling, scoring, policy gradient update

→ Training loop: rollout → Reward → Advantage → PPO update

2. DPO: (Direct Preference Optimization)

Laurel Gemma / Gemini / GPT

can we get rid of PPO?

→ no RL algorithm/no policy model → no reward model

→ works directly on pairwise human preference data. (preferred, rejected)

→ fast/scalable/cheap - no RL loop = KTO, ORPO, SimPO

⇒ Things to watch out for RLHF

→ over-optimization / over-fitting on the reward

→ Mode collapse / entropy = no longer a probabilistic model

Summary: 1. RLHF data collection is hard! Many confounding factors

2. RLHF algorithms are bit more complex than SFT

3. Be mindful of the impact of over-optimizing rewards esp. PPO